# No Szöllősi matrix lies in a quadruplet of mutually unbiased bases in dimension six: a certificate replayed by an independent checker, with a rigorous re-proof for the Fourier family

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## Abstract

Matolcsi, Matszangosz, Varga and Weiner (MMVW, 2026) reduced Zauner's conjecture — that $\mathbb{C}^6$ carries at most three mutually unbiased bases (MUBs) — to an algebraic identity for MUB triplets (their Conjecture 2) together with their Conjecture 3: no complex Hadamard matrix in Szöllősi's two-circulant family $X(\alpha)$ is part of a quadruplet of MUBs. They remarked that Conjecture 3 should be provable with the discretisation technique of Jaming, Matolcsi, Móra, Szöllősi and Weiner (2009), "although the actual implementation could be tedious". We prove Conjecture 3 for the whole closed parameter region, boundary and degenerate loci included.

The proof is a computer-assisted certificate with a deliberately split architecture. An untrusted *generator* covers a fundamental domain of the family's symmetry group by 14,658 parameter boxes and, for each box, produces a finite family of phase-space boxes ("hulls") intended to contain every vector unbiased to both the standard basis and the Hadamard basis, for every parameter in the box. A separate *checker*, which shares no code with the generator, then replays every certificate: its trusted kernel (638 lines of C) moves every floating-point result one unit in the last place outward, uses no library transcendental function, and decides by exhaustive branch and bound (with a modulus test, a mean-value test and a parametric Krawczyk test) that the hulls cover all unbiased vectors; it then recomputes the orthogonality and unbiasedness graphs of the hulls and shows, by its own clique search, that no two further mutually unbiased bases fit into them. The checker also re-derives the parameter cover with its own selector. All 14,658 certificates were verified, 75 of them after subdividing their parameter boxes.

The same interface was applied to the two-parameter Fourier family $F(a, b)$. Exact symmetry and tiling checks justify its fundamental triangle, and 1,935 certificates give a new implementation of the exclusion already proved by Jaming et al. (2009), not a new Fourier exclusion theorem. The general dimension-six maximum remains conditional on MMVW's Conjecture 2. "Independent checker" refers to separation from the generator, not external peer review or proof-assistant formalisation.

## 1. Introduction

### 1.1 The problem

Two orthonormal bases $\{e_k\}$ and $\{f_l\}$ of $\mathbb{C}^d$ are *mutually unbiased* if $|\langle e_k, f_l\rangle|^2 = 1/d$ for all $k, l$. If the bases are the columns of unitary matrices $X_1, X_2$, the condition says that $\sqrt{d}\, X_1^* X_2$ is a *complex Hadamard matrix*: a matrix with unimodular entries whose columns are orthogonal. In prime-power dimensions there are $d + 1$ mutually unbiased bases; in dimension six the largest known sets have three members, and Zauner's conjecture (Zauner 2011) asserts that three is the

maximum. The question is the best-known open case of the MUB problem and has resisted analytic, numerical and algebraic attacks for more than two decades (McNulty and Weigert 2026 give a recent review).

By a unitary change of basis any set of MUBs can be brought into a form in which the first basis is the standard basis ($X_1 = I$) and the second is $H/\sqrt{6}$ for a complex Hadamard matrix $H$. Every further basis consists of unit vectors unbiased to both. Up to a phase, the vectors unbiased to $I$ are $v(\theta)/\sqrt{6}$ with $v(\theta) = (1, e^{i\theta_2}, \dots, e^{i\theta_6})^{\mathsf{T}}$, and unbiasedness to $H/\sqrt{6}$ becomes a system of trigonometric equations in $\theta \in \mathbb{T}^5$ whose solution set we denote by $\mathcal{U}(H)$. A quadruplet $\{I, H/\sqrt{6}, C, D\}$ exists exactly when $\mathcal{U}(H)$ contains two orthonormal bases that are unbiased to each other.

MMVW (2026) studied MUB triplets through their transition matrices and showed that one algebraic identity (their Conjecture 2) forces every Hadamard matrix occurring in a triplet to lie in one of two families: the two-parameter Fourier family $F(a, b)$, which Jaming et al. (2009) had shown cannot be extended to a quadruplet, and Szöllősi's two-parameter family $X(\alpha)$ (Szöllősi 2010). For the latter they stated:

> **Conjecture 3 (MMVW).** Any matrix $S \in X(\alpha)$ in the Szöllősi family, defined in (4.2), cannot be part of an MUB-quadruplet. In principle, this conjecture can be proven with the discretization technique in [Jaming et al. 2009], although the actual implementation could be tedious.

An earlier claim of this statement, derived from a lemma of Chen, Liang, Hu and Chen (2021), is not secure: McNulty and Weigert (2025) identified a defect in the underlying argument, and the reply of Liang and Chen (2025) does not restore it. The display (4.2) of MMVW is Szöllősi's two-circulant form

$$S = \begin{bmatrix} A & B \\ B^* & -A^* \end{bmatrix}, \qquad A = \mathrm{circ}(a, b, c), \quad B = \mathrm{circ}(d, e, f), \qquad |a| = \cdots = |f| = 1,$$

and $X(\alpha)$ denotes the complex Hadamard matrices of this form, which up to equivalence are parametrised by a complex number $\alpha$ in a closed region $\mathbb{D}$ bounded by arcs of a deltoid and its negative (Section 3).

### 1.2 Results

> **Theorem 1.** Let $S$ be a complex Hadamard matrix of order six of the form (4.2). There are no unitary matrices $C, D$ such that $\{I, S/\sqrt{6}, C, D\}$ is a set of four mutually unbiased bases of $\mathbb{C}^6$. Equivalently, no member of $X(\alpha)$, for any $\alpha \in \mathbb{D}$ including the boundary, is part of an MUB quadruplet.

> **Corollary 2.** If Conjecture 2 of MMVW holds, then the maximal number of mutually unbiased bases in $\mathbb{C}^6$ is three.

The corollary is Proposition 4.2 of MMVW with Conjecture 3 discharged by Theorem 1; the Fourier half of their argument rests on Jaming et al. (2009), which we re-prove:

> **Theorem 3 (Jaming, Matolcsi, Móra, Szöllősi and Weiner 2009, Theorem 1.4).** For no $(a, b) \in [0, 1)^2$ can the pair $\{I, F(a, b)\}$ be extended to a quadruplet of mutually unbiased bases.

Theorem 3 is not new. We supply a separated checker and inspectable certificates for the published Fourier exclusion. No implementation-specific criticism of the historical software is needed for this contribution. Fourier also tests transfer between the two implemented families: they share a checker and certificate format, with family-specific exponent data, coefficient rectangles and selectors. Extensions beyond these families remain future work.

### 1.3 What is new, and how it relates to Jaming et al.

The overall strategy — cover the parameter space by cells, enclose the unbiased vectors for each cell, and exclude quadruplets combinatorially — is that of Jaming et al. (2009), and MMVW explicitly anticipated its use for Conjecture 3. The contributions of this paper are the following.

1. *The theorem.* Theorem 1 is new; it removes one of the two remaining hypotheses in MMVW's route to Zauner's conjecture.
2. *A family-level certificate interface* (Section 2): an *admissible family* is specified by integer exponent vectors, unimodular coefficients and an optional trigonometric constraint; a *certificate* for a parameter box is a list of hulls in the phase torus; and four proof obligations (selector soundness, coverage, self-separation of hulls, and a clique condition) imply, for every fundamental-domain parameter in its ownership cell, that the Hadamard matrix is not part of a quadruplet. Closed boxes supply the interval enclosures; half-open cells supply ownership. The interface is what made the Fourier family a change of input data rather than a new program.
3. *Treatment of degenerate parameters without isolating roots.* Along the ray $\arg\alpha = \pi/6$ and near $\alpha = 0$ unbiased vectors coalesce (Section 10). Jaming et al. separated vectors with a fixed-resolution grid; methods that isolate simple roots fail near coalescence at any fixed resolution. Here the exclusion of phase boxes is driven by a *parametric* Krawczyk operator acting uniformly over a parameter box, and hulls are allowed to contain clusters of solutions; the combinatorial step only needs the hulls to be small enough for the self-separation condition.
4. *A verification design with an explicit trusted computing base* (Sections 5–8). The generator is untrusted. The checker re-derives the owner cover, re-verifies phase-space coverage, and recomputes the graph exclusion using outward-rounded intervals. The test suite detects 16 of 17 supplied kernel mutations; that is substantial but incomplete mutation coverage. Section 8 states the surviving mutation and the additional exact generic mixed-term fixture without claiming exhaustive testing.

### 1.4 History of this work

The supplied first review reproduced counterexamples to the earlier generator's radius arithmetic, cyclic hulls and resumption logic, and identified an uncertified selector margin. The version-2 package introduced a separate checker and new certificates. The supplied second review then identified the closed-box ownership mismatch and an input-unbound replay cache. This revision repairs those contracts and withdraws the unused generator-rounding argument. Neither review demonstrated a false exclusion theorem. Review provenance is not independently authenticated, and no external human peer review is asserted. Appendix B records the second-review dispositions; historical records remain distinct from the new clean replay.

### 1.5 Organisation

Section 2 sets up admissible families, the quadruplet property and the certificate, and states the proof obligations. Sections 3 and 4 treat the Szöllősi and Fourier families: parametrisation, exact symmetries, fundamental domains and selectors. Section 5 describes the checker's arithmetic and Section 6 its exclusion tests and coverage verification. Section 7 describes the generator. Section 8 describes the verification architecture, the trusted computing base and the tests. Section 9 reports the computations and Section 10 some numerical diagnostics of degenerate parameters, which are not part of the proof. Section 11 discusses scope and limitations.

## 2. Families, certificates and proof obligations

### 2.1 Admissible families

Write $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$. An *admissible family* of order six with $m$ parameters consists of

- integer exponent vectors $E_{kj} \in \mathbb{Z}^m$ and unimodular constants $c_{kj}$ $(1 \le k, j \le 6)$, defining
$$H(p)_{kj} = c_{kj}\, e^{i\langle E_{kj}, p\rangle}, \qquad p \in \mathbb{T}^m;$$
- optionally, a *constraint* $C(p) = \sum_t \gamma_t\, e^{i\langle e_t, p\rangle}$ with integer vectors $e_t$ and constants $\gamma_t$, whose zero set $\Sigma = \{p : C(p) = 0\}$ is the set of parameters at which $H(p)$ is a complex Hadamard matrix (without a constraint, $\Sigma = \mathbb{T}^m$ and $H(p)$ must be Hadamard for all $p$);
- a *fundamental domain* $\mathcal{F} \subseteq \Sigma$ with the property that the quadruplet question for every $p \in \Sigma$ reduces to the same question for some point of $\mathcal{F}$ (Sections 3.3 and 4.3), together with a *selector* that retains the canonical half-open dyadic owner of every point of $\mathcal{F}$ at every relevant level.

Use representatives in $[0, 2\pi)^m$ and define
$$Q_{\ell,i} = \prod_r [2\pi i_r/2^\ell, 2\pi(i_r+1)/2^\ell), \qquad P_{\ell,i} = \overline{Q_{\ell,i}}.$$
Closure is on the torus; the endpoint $2\pi$ is represented by 0 for ownership. Each point has a unique owner at each level, and its finer owner is a child of its coarser owner. Closed enclosures are used for interval calculations. A discarded enclosure may meet $\mathcal{F}$ on a shared boundary.

For the Szöllősi family $m = 4$, all $c_{kj} = \pm 1$, and the constraint has six terms (Section 3). For the Fourier family $m = 2$, the $c_{kj}$ are sixth roots of unity, and there is no constraint (Section 4). The checker consumes exactly this data: the exponent arrays, rectangles enclosing the constants, and the selector.

### 2.2 The quadruplet property

For a complex Hadamard matrix $H$ of order six let $\mathcal{E}(H)$ be the statement: *there exist unitary $C, D$ such that $\{I, H/\sqrt{6}, C, D\}$ is a set of four mutually unbiased bases.* Two Hadamard matrices are *equivalent* if $H' = M_1 H M_2$ with monomial unitary matrices $M_1, M_2$ (products of a permutation matrix and a diagonal unitary matrix).

**Lemma 4.** (i) If $H'$ is equivalent to $H$ then $\mathcal{E}(H) \iff \mathcal{E}(H')$. (ii) $\mathcal{E}(H) \iff \mathcal{E}(\overline{H})$. (iii) $\mathcal{E}(H) \iff \mathcal{E}(H^*)$, hence $\mathcal{E}(H) \iff \mathcal{E}(H^{\mathsf{T}})$.

*Proof.* (i) Let $\{I, M_1 H M_2/\sqrt{6}, C, D\}$ be a quadruplet. Multiplying all four unitaries on the left by $M_1^{-1}$ preserves mutual unbiasedness and gives $\{M_1^{-1}, H M_2/\sqrt{6}, M_1^{-1}C, M_1^{-1}D\}$. The columns of the monomial unitary $M_1^{-1}$ are the standard basis vectors up to phases, and the columns of $HM_2$ are those of $H$ up to order and phases; as bases these are $I$ and $H/\sqrt{6}$. The converse is the same argument with $M_1^{-1}, M_2^{-1}$. (ii) Conjugate all four matrices; conjugation is an involution. (iii) Put $U = H/\sqrt{6}$ and let $\{I, U, C, D\}$ be a quadruplet. Then $\{I, U^*, U^*C, U^*D\}$ is one: $U^*$, $U^*C$, $U^*D$ have entries of modulus $1/\sqrt{6}$ because $U$ is unbiased to $I$, $C$ and $D$; $(U^*)^*(U^*C) = C$ and $(U^*)^*(U^*D) = D$ are unbiased to $I$; and $(U^*C)^*(U^*D) = C^*D$ is unbiased because $C$ and $D$ are. Since $(H^*)^* = H$ this is an equivalence, and $H^{\mathsf{T}} = \overline{H^*}$ with (ii) gives the last claim. □

### 2.3 Unbiased vectors

For a complex Hadamard matrix $H$ and $\theta = (\theta_2, \dots, \theta_6) \in \mathbb{T}^5$ put $v(\theta) = (1, e^{i\theta_2}, \dots, e^{i\theta_6})^{\mathsf{T}}$ and
$$S_j(\theta) = \left(H^* v(\theta)\right)_j = \sum_{k=1}^{6} \overline{H_{kj}}\, v_k(\theta), \qquad j = 1, \dots, 6.$$

The unit vector $v(\theta)/\sqrt{6}$ is unbiased to the columns of $H/\sqrt{6}$ if and only if $|S_j(\theta)|^2 = 6$ for all $j$. Let

$$\mathcal{U}(H) = \{\theta \in \mathbb{T}^5 : \ |S_j(\theta)|^2 = 6, \ j = 1, \dots, 6\}.$$

(For a Hadamard matrix $\sum_j |S_j|^2 = \|H^* v\|^2 = 36$, so one of the six equations is implied by the others; the tests of Section 6 use either all six or five of them, which is legitimate because a necessary condition for membership in $\mathcal{U}(H)$ is used to exclude.) No finiteness of $\mathcal{U}(H)$ is assumed anywhere.

**Lemma 5.** If $\mathcal{E}(H)$ holds, there are twelve distinct points $\theta^{(1)}, \dots, \theta^{(12)} \in \mathcal{U}(H)$ such that $\langle v(\theta^{(a)}), v(\theta^{(b)})\rangle = 0$ for $a \neq b$ both in $\{1, \dots, 6\}$ or both in $\{7, \dots, 12\}$, and $|\langle v(\theta^{(a)}), v(\theta^{(b)})\rangle|^2 = 6$ for $a \leq 6 < b$.

*Proof.* Each column of $C$ and of $D$ is unbiased to $I$, so all its entries have modulus $1/\sqrt{6}$. Multiplying a column by a unimodular phase changes neither orthogonality within its basis nor unbiasedness to other bases; after such a phase and the factor $\sqrt{6}$ the column is $v(\theta)$ with $\theta \in \mathcal{U}(H)$. Orthogonality within $C$ and $D$ gives the zero inner products, and unbiasedness of $C$ and $D$, $|\langle c_a, d_b\rangle|^2 = 1/6$ for unit vectors, becomes $|\langle v, w\rangle|^2 = 6$ after rescaling by $\sqrt{6}$ each. Two of the twelve points coincide only if $|\langle v, v\rangle|^2 = 36$ equals 0 or 6, which is impossible. □

### 2.4 Hulls and the combinatorial certificate

A *hull* is a box $\prod_{k=2}^{6} [l_k, h_k] \cdot \frac{2\pi}{2^{16}}$ in the phase torus with integer end-points $l_k \leq h_k$ (if $h_k - l_k \geq 2^{16}$ the factor is the whole circle). For hulls $\mathsf{h}, \mathsf{h}'$ let $\Pi(\mathsf{h}, \mathsf{h}') \subseteq [0, \infty)$ be an interval containing $|\langle v(\theta), v(\phi)\rangle|^2$ for all $\theta \in \mathsf{h}$, $\phi \in \mathsf{h}'$, and $\Pi(\mathsf{h}, \mathsf{h})$ one containing these values for all $\theta, \phi \in \mathsf{h}$. Define two graphs on a finite family $\mathcal{H}$ of hulls:

$$\mathsf{h} \sim_O \mathsf{h}' \iff \mathsf{h} \neq \mathsf{h}', \ 0 \in \Pi(\mathsf{h}, \mathsf{h}'), \qquad \mathsf{h} \sim_B \mathsf{h}' \iff \mathsf{h} \neq \mathsf{h}', \ 6 \in \Pi(\mathsf{h}, \mathsf{h}').$$

**Proposition 6 (combinatorial certificate).** Let $H$ be a complex Hadamard matrix and $\mathcal{H}$ a finite family of hulls such that

- (Cov) $\mathcal{U}(H) \subseteq \bigcup_{\mathsf{h} \in \mathcal{H}} \mathsf{h}$;
- (Sep) $\Pi(\mathsf{h}, \mathsf{h}) \subset (6, \infty)$ for every $\mathsf{h} \in \mathcal{H}$;
- (Cl) there are no two six-cliques $K_1, K_2$ of the graph $O$ such that every pair in $K_1 \times K_2$ is an edge of $B$.

Then $\mathcal{E}(H)$ fails.

*Proof.* Suppose $\mathcal{E}(H)$ holds, take the twelve points of Lemma 5 and assign to each a hull containing it (Cov). For two distinct points of the twelve, $|\langle v, w\rangle|^2 \in \{0, 6\}$; by (Sep) they cannot share a hull, so the twelve hulls are distinct. The six hulls of the first basis are pairwise joined in $O$, likewise those of the second basis, and every cross pair is joined in $B$ (by the definition of $\Pi$). This contradicts (Cl). □

The certificate is *uniform in the parameter*: if (Cov) holds for $H(p)$ for every $p$ in a set $P'$, and (Sep) and (Cl) hold for the hull family (they do not involve $p$), then $\mathcal{E}(H(p))$ fails for every $p \in P'$.

### 2.5 Proof obligations

A *certificate* for a closed dyadic parameter box $P$ is a finite hull family $\mathcal{H}_P$. A *cover* is a set of certified boxes. The following statement is what the checker establishes.

**Theorem 7.** Let an admissible family with fundamental domain $\mathcal{F}$ and selector be given, and a cover with certificates $\mathcal{H}_P$. Suppose

- (O1) *selector soundness and cover completeness:* every kept cell of the selector's cover tree (Section 6.5) is either certified or has all its kept children covered, down to a finite depth; every $p \in \mathcal{F}$ has its canonical owner retained at each relevant level;
- (O2) *coverage:* for every certified ownership cell $Q$ with enclosure $P$ and every $p \in \mathcal{F} \cap Q$, $\mathcal{U}(H(p)) \subseteq \bigcup \mathcal{H}_P$;
- (O3) *separation:* (Sep) holds for every hull of every $\mathcal{H}_P$;
- (O4) *clique condition:* (Cl) holds for every $\mathcal{H}_P$.

Then $\mathcal{E}(H(p))$ fails for every $p \in \mathcal{F}$, and hence, by the defining property of $\mathcal{F}$, for every $p \in \Sigma$.

*Proof.* Follow the unique owners of $p \in \mathcal{F}$ from the top level. By (O1) they are retained, nested, and reach a certified cell $Q$ at finite depth. Thus $p \in \mathcal{F} \cap Q$, and (O2)–(O4) give Proposition 6 with $\mathcal{H} = \mathcal{H}_P$. □

In the Szöllősi case the reduction from $\Sigma$ to $\mathcal{F}$ is Proposition 10, and every normalised matrix of the form (4.2) is $H(p)$ for some $p \in \Sigma$ (Lemma 8); together these give Theorem 1. In the Fourier case the reduction is Proposition 12, which gives Theorem 3.

# 3. The Szöllősi family

## 3.1 Normal form and parameters

Throughout, $\mathrm{circ}(a_0, a_1, a_2)$ is the $3 \times 3$ circulant with rows $(a_0, a_1, a_2)$, $(a_2, a_0, a_1)$, $(a_1, a_2, a_0)$. Circulants commute, are normal, and $\mathrm{circ}(a_0, a_1, a_2)^{\mathsf{T}} = \mathrm{circ}(a_0, a_2, a_1)$.

**Lemma 8 (normalisation and the Hadamard condition).** (i) Multiplying rows 1–3 of $S$ in (4.2) by $\lambda$ and rows 4–6 by $\bar{\lambda}$, or columns 1–3 by $\mu$ and columns 4–6 by $\bar{\mu}$ ($|\lambda| = |\mu| = 1$), gives again a matrix of the form (4.2), with $(A, B)$ replaced by $(\lambda\mu A, \lambda\bar{\mu}B)$; consequently every matrix of the form (4.2) is equivalent to one with $a = d = 1$. (ii) A matrix of the form (4.2) with $a = d = 1$ is a complex Hadamard matrix if and only if $\bar{b} + b\bar{c} + c + \bar{e} + e\bar{f} + f = 0$.

*Proof.* (i) The scaled matrix is $\begin{bmatrix} \lambda\mu A & \lambda\bar{\mu}B \\ \bar{\lambda}\mu B^* & -\bar{\lambda}\bar{\mu}A^* \end{bmatrix}$, and $(\lambda\bar{\mu}B)^* = \bar{\lambda}\mu B^*$, $(\lambda\mu A)^* = \bar{\lambda}\bar{\mu}A^*$. Choosing $\lambda^2 = \bar{a}\bar{d}$ and $\mu = \bar{a}/\lambda$ gives $\lambda\mu = \bar{a}$, $\lambda\bar{\mu} = \bar{d}$. (ii) Since $A, B$ are circulant, $SS^* = \mathrm{diag}(AA^* + BB^*,\ B^*B + A^*A)$ and $A^*A = AA^*$, $B^*B = BB^*$. $AA^*$ is the circulant whose first row is the autocorrelation of $(1, b, c)$, namely $(3,\ \bar{b}+b\bar{c}+c,\ \overline{\bar{b} + b\bar{c} + c})$, and likewise for $B$. Hence $SS^* = 6I$ if and only if the stated sum vanishes. □

For $p = (\theta_x, \theta_y, \theta_u, \theta_v) \in \mathbb{T}^4$ put

$$x = e^{i\theta_x},\ y = e^{i\theta_y},\ z = \overline{xy}, \qquad u = e^{i\theta_u},\ v = e^{i\theta_v},\ w = \overline{uv},$$

$$A(p) = \mathrm{circ}(1, \bar{x}, z), \quad B(p) = \mathrm{circ}(1, \bar{u}, w), \qquad H(p) = \begin{bmatrix} A(p) & B(p) \\ B(p)^* & -A(p)^* \end{bmatrix}.$$

This is (4.2) with $a = d = 1$, $b = \bar{x}$, $c = z$, $e = \bar{u}$, $f = w$, and every unimodular $(b, c, e, f)$ arises ($x = \bar{b}$, $y = b\bar{c}$, $u = \bar{e}$, $v = e\bar{f}$). In these variables $\bar{b} + b\bar{c} + c = x + y + z$, so by Lemma 8 $H(p)$ is Hadamard if and only if

$$C(p) := x + y + z + u + v + w = 0. \tag{C}$$

The family is admissible in the sense of Section 2.1: every entry of $H(p)$ is $\pm e^{i\langle E, p\rangle}$ with $E \in \{-1, 0, 1\}^4$ having at most two non-zero entries, and $C$ has six terms. An exact symbolic computation (`exact/lemma5_exact.py`, SymPy, treating $x, y, u, v$ as Laurent variables with $\bar{x} = 1/x$) confirms that every diagonal entry of $H(p)H(p)^*$ is 6 and every off-diagonal entry is 0, $C(p)$ or $\overline{C(p)}$, identically in $p$.

*Relation to Szöllősi's parameter.* Put $\alpha = x + y + z$. Since $xyz = 1$ and $|x| = |y| = |z| = 1$, $xy + yz + zx = \bar{\alpha}$, so $x, y, z$ are the roots of $f_\alpha(t) = t^3 - \alpha t^2 + \bar{\alpha} t - 1$, and on $\Sigma$ the numbers $u, v, w$ are the roots of $f_{-\alpha}$. Szöllősi's region is $\mathbb{D} = \{\alpha : D[\alpha] \le 0,\ D[-\alpha] \le 0\}$ with $D[\alpha] = |\alpha|^4 + 18|\alpha|^2 - 8\,\mathrm{Re}(\alpha^3) - 27$, the condition for $f_\alpha$ to have three unimodular roots (Szöllősi 2010, Section 3). Conversely, for $\alpha \in \mathbb{D}$ every labelling of the roots of $f_\alpha$ as $(x, y, z)$ and of $f_{-\alpha}$ as $(u, v, w)$ gives a point of $\Sigma$. Thus $\Sigma$ is precisely the set of normalised matrices of the family for all $\alpha$ in the *closed* region $\mathbb{D}$ and all labellings; points on the boundary of $\mathbb{D}$, where roots coincide, are points of $\Sigma$ like any others and need no separate treatment.

### 3.2 Symmetries

Let $\Pi = \mathrm{circ}(0, 1, 0)$, let $P_{23}$ exchange indices 2 and 3, $D_\omega = \mathrm{diag}(1, \omega, \omega^2)$ with $\omega = e^{2\pi i/3}$, and write $M_1 \oplus M_2$ for a block-diagonal matrix. Define the following maps of $\mathbb{T}^4$, written through the six unimodular numbers $(x, y, z, u, v, w)$, which determine $p$:

$$\begin{aligned}
\sigma_1 &: (x, y, z, u, v, w) \mapsto (y, z, x, u, v, w), & \sigma_2 &: (x, y, z, u, v, w) \mapsto (x, y, z, v, w, u),\\
\rho &: (x, y, z, u, v, w) \mapsto (\bar{z}, \bar{y}, \bar{x}, \bar{w}, \bar{v}, \bar{u}), & \tau &: (x, y, z, u, v, w) \mapsto (\bar{z}, \bar{y}, \bar{x}, \bar{u}, \bar{v}, \bar{w}),\\
\kappa &: (x, y, z, u, v, w) \mapsto (\bar{x}, \bar{y}, \bar{z}, \bar{u}, \bar{v}, \bar{w}), & \omega_\cdot &: (x, y, z, u, v, w) \mapsto \omega\,(x, y, z, u, v, w),\\
\varsigma &: (x, y, z, u, v, w) \mapsto (u, v, w, x, y, z).
\end{aligned}$$

**Lemma 9 (matrix identities).** For every $p \in \mathbb{T}^4$:

1. $H(\tau p) = \overline{H(p)}^{\mathsf{T}}$;
2. $H(\kappa p) = \overline{H(p)}$;
3. $H(\rho p) = (P_{23} \oplus P_{23})\, H(p)\, (P_{23} \oplus P_{23})$;
4. $H(\omega_\cdot p) = (D_\omega \oplus D_\omega)\, H(p)\, (D_\omega \oplus D_\omega)^{-1}$;
5. $H(\varsigma p) = (I \oplus (-I))\, H(p) \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$;
6. $H(\sigma_1 p) = (xI \oplus \Pi)\, H(p)\, (\Pi^{\mathsf{T}} \oplus \bar{x} I)$ and $H(\sigma_2 p) = (uI \oplus \Pi)\, H(p)\, (\bar{u} I \oplus \Pi^{\mathsf{T}})$.

Moreover $C(\sigma_1 p) = C(\sigma_2 p) = C(\varsigma p) = C(p)$, $C(\rho p) = C(\tau p) = C(\kappa p) = \overline{C(p)}$ and $C(\omega_\cdot p) = \omega C(p)$, so every map preserves $\Sigma$; and $\alpha$ is mapped to $\alpha$ by $\sigma_1, \sigma_2$, to $\bar{\alpha}$ by $\rho, \tau, \kappa$, to $\omega\alpha$ by $\omega_\cdot$, and to $u + v + w$ ($= -\alpha$ on $\Sigma$) by $\varsigma$.

*Proof.* Write $A = \mathrm{circ}(1, \bar{x}, z)$, $B = \mathrm{circ}(1, \bar{u}, w)$. (1) $H^{\mathsf{T}} = \begin{bmatrix} A^{\mathsf{T}} & \bar{B} \\ B^{\mathsf{T}} & -\bar{A} \end{bmatrix}$ with $A^{\mathsf{T}} = \mathrm{circ}(1, z, \bar{x})$ and $\bar{B} = \mathrm{circ}(1, u, \bar{w})$; this pair is $(\mathrm{circ}(1, \bar{x}', z'), \mathrm{circ}(1, \bar{u}', w'))$ with $x' = \bar{z}$, $z' = \bar{x}$, $y' = \bar{y}$, $u' = \bar{u}$, $w' = \bar{w}$, $v' = \bar{v}$, and the lower blocks are the required adjoints. (2) is immediate. (3) Conjugation by $P_{23}$ transposes a circulant; the rest is as in (1). (4) $D_\omega\, \mathrm{circ}(a_0, a_1, a_2)\, D_\omega^{-1} = \mathrm{circ}(a_0, \bar{\omega} a_1, \bar{\omega}^2 a_2)$. (5) $H \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} = \begin{bmatrix} B & A \\ -A^* & B^* \end{bmatrix}$, and negating the last three rows gives $H(\varsigma p)$. (6) $xA = \mathrm{circ}(x, 1, xz) = \mathrm{circ}(x, 1, \bar{y})$ is a cyclic shift of $A(\sigma_1 p) = \mathrm{circ}(1, \bar{y}, x)$; the shifts commute with circulants and cancel on the lower-right block. All identities, including the exponent signs in (6), which the earlier version left as "a suitable choice", and the statements about $C$ and $\alpha$, were verified by exact symbolic computation (`exact/lemma5_exact.py`, output `lemma5_exact.json`). □

By Lemma 4, $\mathcal{E}(H(gp)) \iff \mathcal{E}(H(p))$ for every $g$ in the group $G$ generated by the seven maps.

### 3.3 A fundamental domain

Use angle coordinates $\theta_x, \ldots, \theta_w \in [0, 2\pi)$ for the six numbers, with $\theta_z \equiv -\theta_x - \theta_y$ and $\theta_w \equiv -\theta_u - \theta_v$. Consider

- (S) $\alpha = x + y + z$ lies in the closed sector $\{0 \le \arg\alpha \le \pi/6\} \cup \{0\}$;

- (OA) $\theta_x \le \theta_y \le \theta_z$; (OB) $\theta_u \le \theta_v \le \theta_w$.

**Proposition 10.** For every $p \in \Sigma$ there is $g \in G$ such that $gp$ satisfies (S), (OA) and (OB). Hence if $\mathcal{E}(H(p))$ fails for every $p$ in $\mathcal{F} := \{p \in \mathbb{T}^4 : \text{(C)}, \text{(S)}, \text{(OA)}, \text{(OB)}\}$, it fails for every $p \in \Sigma$.

*Proof.* $\sigma_1$ and $\sigma_2$ permute the labels of the two triples cyclically. The composition $\kappa\tau$ maps $(x, y, z, u, v, w)$ to $(z, y, x, u, v, w)$ and $\kappa\rho$ maps it to $(z, y, x, w, v, u)$ (both verified symbolically), so $(\kappa\rho)(\kappa\tau)$ reverses the second triple alone. Cyclic permutations and a reversal generate $S_3$; hence $G$ contains $S_3 \times S_3$ acting on the labels, which leaves $\{x, y, z\}$, $\{u, v, w\}$ and $\alpha$ unchanged, and sorting the angles of each triple achieves (OA) and (OB). The maps $\omega_\cdot$, $\varsigma$ and $\kappa$ send $\alpha$ to $\omega\alpha$, $-\alpha$ and $\bar{\alpha}$; they generate the dihedral group of order twelve (rotations by multiples of $\pi/3$ and a reflection), whose closed fundamental sector is $0 \le \arg\alpha \le \pi/6$ ($\alpha = 0$ is fixed). After moving $\alpha$ into the sector, a relabelling, which does not change $\alpha$, restores (OA) and (OB). All maps preserve (C). □

### 3.4 The selector

Ownership cells and their closed enclosures are as in Section 2.1. The enclosure indexed by $i$ at level $\ell$ has centre $c = (2i+1)\pi/2^\ell$ and half-width $h = \pi/2^\ell$. Centres lie on $2\pi\mathbb{Z}/2^{17}$ for $\ell \le 16$. The selector rejects only when a test excludes the ownership cell from $\mathcal{F}$; the ordering test need not exclude its entire closed enclosure.

*Ordering (OA), (OB): exact integer test.* In units of $2\pi/2^\ell$ write $t_1 \in [i_1, i_1+1)$, $t_2 \in [i_2, i_2+1)$ for $\theta_x, \theta_y$. Every point of $\mathcal{F}$ has one half-open owner at each level. Direct interval tests use its closed enclosure; when subdivision is needed, only the retained fundamental-domain portion is certified by following the unique ownership path. Then $\theta_z$ corresponds to $t_3 = (-t_1 - t_2) \bmod N$, $N = 2^\ell$. (OA) requires $t_1 \le t_2 \le t_3$, hence $i_1 \le i_2$ and $\sup t_3 \ge i_2$. The quantity $-(t_1+t_2)$ ranges over $(\beta-2, \beta]$ with $\beta = -(i_1+i_2)$, which contains the integers $\beta-1$ and $\beta$; if one of them is a multiple of $N$ (that is, $\beta \bmod N \in \{0, 1\}$) the residue $t_3$ approaches $N$ and $\sup t_3 = N$, otherwise $\sup t_3 = \beta \bmod N$. The box is discarded if $i_1 > i_2$ or $\sup t_3 < i_2$; (OB) is treated identically. The generator uses an algebraically different formula for the same condition; the checker's test-suite compares its formula with brute-force enumeration on fine rational sub-grids.

*Constraint (C): support-function test with rigorous rounding.* Let $D(c) = (i(x-z), i(y-z), i(u-w), i(v-w))$ be the gradient of $C$ at the centre (complex numbers viewed in $\mathbb{R}^2$). For $|\delta_r| \le h$ Taylor's theorem gives $C(c+\delta) = C(c) + \sum_r D_r(c)\delta_r + R$ with $|R| \le 6h^2$: the Hessian of each term $e^{i\langle e_t, p\rangle}$ in direction $\delta$ has modulus at most $\langle e_t, \delta\rangle^2 \le \|e_t\|_1^2 h^2$, and $\sum_t \|e_t\|_1^2 = 1+1+4+1+1+4 = 12$, so the second-order remainder is at most $\frac{1}{2} \cdot 12h^2$. If $C(c+\delta) = 0$ then for every vector $n \in \mathbb{R}^2$

$$\langle n, -C(c)\rangle \le h \sum_r |\langle n, D_r(c)\rangle| + 6h^2|n|.$$

The box is discarded if some $n$ violates this inequality. The candidate vectors $n$ (edge normals and vertex directions of the zonotope $\{\sum_r D_r\delta_r\}$ and $-C(c)$ itself, 25 in all) are computed in ordinary floating point, which is harmless because the inequality holds for *every* $n$; what must be rigorous is its evaluation, and in the checker every term is evaluated in the interval arithmetic of Section 5: $C(c)$ and $D_r(c)$ are rectangles built from verified table entries, $|n|$ is bounded above with outward steps, $h \ge \pi/2^\ell$ is an upward-rounded product with an upper bound of $\pi$, and the comparison is between a rigorous lower bound of the left side and a rigorous upper bound of the right side.

*Sector (S).* Over the box, $|x-x(c)|, |y-y(c)| \le h$ and $|z-z(c)| \le 2h$ (chord at most arc), so $\alpha$ lies within $4h$ of $\alpha(c)$, which lies in a rigorously computed rectangle. The closed sector is contained in each of the half-planes $\{\operatorname{Im}\alpha \ge 0\}$, $\{\operatorname{Re}\alpha \ge 0\}$ and $\{\operatorname{Re}\alpha - \sqrt{3}\operatorname{Im}\alpha \ge 0\}$. The box is discarded

if the disc of radius $4h$ about the rectangle lies strictly outside one of them: $\max \mathrm{Im} + 4h < 0$, or $\max \mathrm{Re} + 4h < 0$, or $\min(\sqrt{3}\,\mathrm{Im} - \mathrm{Re}) > 2 \cdot 4h$ (the normal $(-1, \sqrt{3})$ has length 2), with $\sqrt{3}$ replaced by the neighbouring binary64 numbers below and above it (verified with 60-digit arithmetic) according to the sign of Im.

**Lemma 11 (owner retention).** The selector retains the canonical owner of every $p \in \mathcal{F}$ at each relevant level. A discarded half-open ownership cell does not meet $\mathcal{F}$; its closed enclosure may do so.

*Proof.* The ordering inequalities above are necessary for the half-open owner of a point satisfying (OA)/(OB). The constraint and sector tests exclude only when their condition fails throughout the closed enclosure, hence throughout the ownership cell. A point in $\mathcal{F}$ fails none of these necessary conditions in its owner. □

At level 9, take $t = \pi/32$, $u = v = -e^{it}$, $w = e^{-2it}$, $\alpha = 2e^{it} - e^{-2it}$ and ordered roots $x, y, z$ of $f_\alpha$. The resulting fundamental-domain point has retained owner $(34, 106, 264, 264)$. It also lies on the upper $v$ boundary of the discarded closed enclosure $(34, 106, 264, 263)$. This permanent regression illustrates why closed-enclosure retention is not asserted.

These tests replace the selector of the earlier version, which evaluated centres with the library exponential and relied on a margin of $10^{-9}$ (referee comment 4.2). Starting from level 3, the checker's selector keeps 160, 183, 232, 532, 1,367, 4,294 and 14,658 boxes at levels $3, 4, \ldots, 9$. The level-9 boxes are the ones certified; the 1,367 kept level-7 boxes ("roots") organise the computation (Section 7).

# 4. The Fourier family

## 4.1 Definition

Following Jaming et al. (2009, eq. (3)), with $\omega = e^{2\pi i/3}$, $x = e^{2\pi i a}$ and $y = e^{2\pi i b}$,

$$\sqrt{6}\,F(a,b) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -\omega^2 x & \omega & -x & \omega^2 & -\omega x \\ 1 & \omega y & \omega^2 & y & \omega & \omega^2 y \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & \omega^2 x & \omega & x & \omega^2 & \omega x \\ 1 & -\omega y & \omega^2 & -y & \omega & -\omega^2 y \end{bmatrix}, \qquad (a,b) \in [0,1)^2.$$

In the notation of Section 2.1 this is an admissible family with $m = 2$, $p = (2\pi a, 2\pi b)$, exponent vectors $E_{kj} \in \{(0,0), (1,0), (0,1)\}$, coefficients $c_{kj} = \zeta^{z_{kj}}$ with $\zeta = e^{i\pi/3}$, and no constraint. An exact symbolic computation (`fourier/fourier_exact.py`) confirms that $\sqrt{6}F(a,b)\,(\sqrt{6}F(a,b))^* = 6I$ identically in the unimodular variables $x, y$. The checker encloses each coefficient $\zeta^z$ in a rectangle computed with `mpmath` interval arithmetic and widened outward by one unit in the last place (the coefficients $\pm 1$ are exact); its test-suite compares the family data with the displayed formula at random parameters.

MMVW display the transposed family $F^{\mathsf{T}}$; by Lemma 4(iii) the quadruplet question is the same for $F$ and $F^{\mathsf{T}}$.

## 4.2 Symmetries and the fundamental triangle

Jaming et al. restrict $(a, b)$ to the triangle $T$ with vertices $(0,0)$, $(1/6, 0)$ and $(1/6, 1/12)$, "due to the equivalences described in [Bengtsson et al. 2007]". To make the re-proof self-contained we derive and verify the needed symmetries directly. Every entry of $\sqrt{6}F(a,b)$ is $\eta^z x^i y^j$ with

$\eta = e^{i\pi/6}$, which we record as an exponent triple $(z \bmod 12, i, j)$. Call an affine map

$$T(a,b) = \Big(pa + qb + \tfrac{m}{12},\ ra + tb + \tfrac{n}{12}\Big), \qquad p,q,r,t,m,n \in \mathbb{Z},\ pt - qr = \pm 1,$$

a *symmetry* if $F(T(a,b)) = D_1 P_1 F(a,b) P_2 D_2$ identically in $(a,b)$, with permutation matrices $P_1, P_2$ and diagonal unitary matrices $D_1, D_2$ depending on $(a,b)$. (Transposition and conjugation were also allowed in the search but were not needed.) In exponent notation this says that for some permutations $\pi, \sigma$ the triples $R_{kj} = E'_{kj} - E_{\pi(k)\sigma(j)}$ — where $E'$ are the exponents of $F(T(a,b))$, obtained from those of $F$ by an integer substitution — have the form $\alpha_k + \beta_j$ in $\mathbb{Z}_{12} \times \mathbb{Z} \times \mathbb{Z}$, and then $D_1 = \mathrm{diag}(\eta^{\alpha_{k,0}} x^{\alpha_{k,1}} y^{\alpha_{k,2}})$ and $D_2$ likewise. This is a finite, exact integer computation.

**Proposition 12.** (i) Among the affine maps with $|p|, |q|, |r|, |t| \le 2$ and $m, n \in \{0, \dots, 11\}$ there are exactly 144 symmetries. They form a group under composition modulo $\mathbb{Z}^2$: the linear parts are the twelve integer matrices preserving the quadratic form $a^2 - ab + b^2$ (the dihedral group of order twelve of the hexagonal lattice; for example $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$), and the translations are the twelve elements of the subgroup of $(\frac{1}{12}\mathbb{Z}/\mathbb{Z})^2$ generated by $(\frac{1}{6}, \frac{1}{3})$ and $(0, \frac{1}{2})$. (ii) The triangle $T$ is a fundamental domain: every $(a,b) \in [0,1)^2$ has an image in $T$ under one of the 144 symmetries. Consequently, if $\mathcal{E}(F(a,b))$ fails for all $(a,b) \in T$, it fails for all $(a,b)$.

*Proof.* (i) is the output of an exhaustive exact search (`fourier/symmetries.py`), which for each of the $104 \cdot 144$ candidate maps first compares an equivalence invariant (the multiset of the exponents of $F_{ki} F_{lj} \overline{F_{kj} F_{li}}$, which is unchanged by row and column permutations and by diagonal scalings) and then, for the 1,296 maps that pass, searches all 720 row permutations for a column permutation and diagonal factors making $R_{kj}$ of rank-one additive form. Every one of the 144 identities was also confirmed numerically at random parameters (maximal deviation $4 \cdot 10^{-15}$), and closure under composition and inversion was checked exactly. (ii) The preimages $T_g^{-1}(T)$ of the triangle under the 144 maps, reduced modulo $\mathbb{Z}^2$, have total area $144 \cdot \frac{1}{144} = 1$ (each map preserves area), and any two of them, including translates by $\mathbb{Z}^2$, have disjoint interiors: for each of the 511,416 pairs a separating line parallel to an edge was found in exact rational arithmetic (`fourier/fundamental_domain.py`). Their union is therefore a closed subset of the torus of full measure, hence the whole torus; so every $(a,b)$ lies in some $T_g^{-1}(T)$, i.e. $T_g(a,b) \in T$. The last claim follows from Lemma 4(i). □

### 4.3 The selector

Parameter boxes are dyadic in $(a,b)$: at level $\ell$ the box $[i, i+1] \times [j, j+1] \cdot 2^{-\ell}$. A point of $T$ with $a \in [i2^{-\ell}, (i+1)2^{-\ell})$ and $b \in [j2^{-\ell}, (j+1)2^{-\ell})$ satisfies $i2^{-\ell} \le a \le \frac{1}{6}$ and $j2^{-\ell} \le b \le a/2 < (i+1)2^{-\ell-1}$, hence $6i \le 2^\ell$ and $2j \le i + 1$. The selector keeps exactly the boxes satisfying these two integer inequalities; it is exact and needs no floating point. It keeps 3, 5, 15, 41, 143, 505 and 1,935 boxes at levels $3, \dots, 9$; the level-9 boxes are certified.

## 5. The checker's arithmetic

### 5.1 One-ulp outward rounding

The checker's kernel (`chk/kernel.c`) performs all floating-point work of the proof. It uses IEEE-754 binary64 arithmetic in the default rounding mode, round-to-nearest, and never changes the mode; the only operations applied to enclosure end-points are $+, -, \times$ and $\sqrt{\ }$. It is compiled without value-changing optimisations (`-O2 -ffp-contract=off -fno-fast-math`), so that no multiply–add fusion or reassociation alters the operations written in the source. For a binary64 number $x$ let $\mathrm{up}(x)$ be the next binary64 number above $x$ and $\mathrm{dn}(x) = -\mathrm{up}(-x)$, computed by integer manipulation of the bit pattern ($\mathrm{up}(0)$ is the smallest positive subnormal, $\mathrm{up}(+\infty) = +\infty$, NaN is propagated).

**Lemma 13.** Let $a, b$ be binary64 numbers, $\circ \in \{+, -, \times\}$, and $s = \mathrm{fl}(a \circ b)$ the round-to-nearest result (possibly $\pm\infty$ after overflow). Then $\mathrm{dn}(s) \le a \circ b \le \mathrm{up}(s)$. The same holds for $s = \mathrm{fl}(\sqrt{a})$, $a \ge 0$.

*Proof.* If $s$ is finite, the exact result lies between the two binary64 neighbours of $s$, because round-to-nearest selects $s$ only for values at most half a spacing away from $s$ on either side, whereas the neighbours are a full spacing away (at a binade boundary the spacing below is half the spacing above, and the half-spacing criterion still refers to the correct side; in the subnormal range the spacing is constant and the argument is the same). If $s = +\infty$ the exact value exceeds the largest finite number $M$, and $\mathrm{dn}(+\infty) = M$; symmetrically for $-\infty$. □

Interval operations are the standard inf–sup rules built from Lemma 13: $[a, b] + [c, d] = [\mathrm{dn}(a + c), \mathrm{up}(b + d)]$, a product takes the minimum and maximum of the four computed end-point products and steps them outward, a product containing $0 \cdot \infty$ becomes $(-\infty, \infty)$, and a square is clamped at 0 from below. Complex quantities are rectangles (pairs of intervals) with the usual rectangular product. Multiplication by an integer exponent $e$ is carried out exactly when $e \in \{0, \pm 1\}$ (the term is omitted or its sign changed) and by an outward-rounded scalar product otherwise; exactly-zero structural entries of a matrix are skipped. None of these choices can make an enclosure smaller than the set it encloses.

The kernel is short enough to be read line by line (Appendix C gives its size and checksum), and its test-suite (Section 8.3) compares every operation with exact rational arithmetic on edge cases (zeros, subnormals, the largest finite number, $1 + 2^{-52}$) and on 4,000 random pairs spread over sixty orders of magnitude.

### 5.2 The unit table and constants

All angles are integers on the grid $2\pi\mathbb{Z}/2^{17}$ ("units", one unit $= A := \pi/2^{16}$). A table $T_m \approx e^{2\pi i m/2^{17}}$, $0 \le m < 2^{17}$, supplies unit complex numbers. The table file is produced by the generator, but the checker does not rely on the generator's verification of it: before any use, the checker evaluates $\cos(2\pi m/2^{17})$ and $\sin(2\pi m/2^{17})$ for every $m$ in `mpmath` interval arithmetic at forty decimal digits and requires the rigorous upper bound of each difference from the stored component to be at most $2^{-53}$. All 262,144 components pass. The kernel then stores the rectangle $[\mathrm{dn}(\mathrm{Re}\, T_m - 2^{-53}), \mathrm{up}(\mathrm{Re}\, T_m + 2^{-53})] \times [\cdots]$, which contains $e^{2\pi i m/2^{17}}$. The worker processes verify the SHA-256 digest of the table file against the digest of the file that was verified.

Three constants enter: an upper bound $\pi^+$ of $\pi$ (the binary64 successor of $\mathrm{fl}(\pi)$), so that $A^+ = \pi^+/2^{16}$ (an exact scaling) bounds $A$ from above; and binary64 numbers $\sqrt{3}^- < \sqrt{3} < \sqrt{3}^+$. All three inequalities are checked in 60-digit arithmetic when the checker starts. The constant 6 is exact.

### 5.3 Enclosures of $e^{i\varphi}$

For an integer centre $c$ and an integer half-width $w \ge 0$ (units), every $\varphi \in [c - w, c + w]A$ satisfies $e^{i\varphi} = e^{icA} e^{it}$ with $|t| \le wA$, and

$$\mathrm{Re}\, e^{it} \in [\cos wA, 1] \ \ (wA \le \pi), \qquad \mathrm{Im}\, e^{it} \in [-\sin wA, \sin wA] \ \ (wA \le \pi/2),$$

with $[-1, 1]$ outside these ranges. The kernel takes $\cos wA$ and $\sin wA$ from the verified rectangle of $T_w$ (the lower end of the real part and the upper end of the imaginary part) and multiplies the rectangle of $T_{c \bmod 2^{17}}$ by the rectangle $[\cos wA, 1] \times [-\sin wA, \sin wA]$. The same enclosure, with $w$ the sum of the half-widths of all angles involved (the phase half-width plus $\sum_r |E_{kjr}|$ times the parameter half-width), gives a rectangle for each term

$$\tau_{kj}(\theta, p) = \overline{c_{kj}}\, e^{i(\theta_k - \langle E_{kj}, p \rangle)}$$

of $S_j = \sum_k \tau_{kj}$ over a phase box and a parameter box, and hence a rectangle for $S_j$. Four variants are used: all angles at their centres; only the free parameters varying; the phases and the dependent parameters varying; and everything varying (Section 6.3). Derivatives are formed from the same rectangles: $\partial_{\theta_k} \tau_{kj} = i\tau_{kj}$ and $\partial_{p_r} \tau_{kj} = -iE_{kjr}\tau_{kj}$, and similarly for second derivatives and for the constraint terms $\gamma_t e^{i\langle e_t, p\rangle}$.

**Lemma 14.** For every phase box $X$, parameter box $P$ and $j$, the rectangle computed for $S_j$ (in any of the four variants) contains $S_j(\theta, p)$ for all $(\theta, p)$ in the corresponding box; likewise for the derivative rectangles.

*Proof.* Lemma 13, the enclosure of $e^{i\varphi}$ above, the table verification, and the inclusion-monotonicity of interval arithmetic. □

# 6. Exclusion tests and the verification of a certificate

Throughout, $X$ is a phase box $\prod_{k=2}^{6} [l_k, h_k] \cdot 2\pi/2^{16}$ (centre $l_k + h_k$ and half-width $h_k - l_k$ in units of $A$), $P$ a closed parameter box with centre $c_P$ and half-width $h_P$, and "$X \times P$ contains no solution" means: there is no $(\theta, p) \in X \times P$ with $p \in \Sigma$ and $\theta \in \mathcal{U}(H(p))$. Only necessary conditions for being a solution are used to exclude, so each test is valid whether or not $H(p)$ is Hadamard at the points of $P$ off $\Sigma$.

## 6.1 Modulus test

For each $j$ the kernel forms the rectangle $M_j$ of $\sum_k \tau_{kj}$ with all angles at their centres, and the radius $R_j = A^+ \sum_k w_{kj}$ with $w_{kj}$ the total half-width of the angle of $\tau_{kj}$ (upward rounded). Since $|e^{i\varphi} - e^{i\psi}| \le |\varphi - \psi|$, $S_j(\theta, p)$ lies within $R_j$ of $M_j$.

**Lemma 15.** If for some $j$ the upper bound of $|M_j| + R_j$ has square below 6, or the lower bound of $\operatorname{dist}(0, M_j) - R_j$ is positive with square above 6, then $X \times P$ contains no solution. □

This test is cheap and discards most of the phase torus; it is also the only test used in the *root pass*, which subdivides the whole phase torus uniformly from level 4 ($2^{20}$ boxes) to level 7 for a level-7 parameter box and keeps the survivors (between $1.8 \cdot 10^6$ and $7.2 \cdot 10^6$ boxes at level 7 for the Szöllősi roots). A box excluded for a parameter box is excluded for every sub-box, so the root pass is shared by all certified boxes below the same root.

## 6.2 Mean-value test

Let $F_j = |S_j|^2 - 6$. The box $X \times P$ is convex, so by the mean-value theorem

$$F_j(\theta, p) \in F_j(\tilde{\theta}, c_P) + \sum_{k=2}^{6} \partial_{\theta_k} F_j(X \times P)\,[-\rho_k, \rho_k] + \sum_r \partial_{p_r} F_j(X \times P)\,[-h_P, h_P],$$

where $\tilde{\theta}$ is the centre of $X$, $\rho_k$ the half-widths of $X$ in radians (upward rounded), and $\partial F_j = 2\,\mathrm{Re}(\overline{S_j}\,\partial S_j)$ is enclosed over the whole box using Lemma 14.

**Lemma 16.** If for some $j$ the interval above does not contain 0, then $X \times P$ contains no solution. □

## 6.3 Parametric Krawczyk test

For a family with a constraint the parameters are split, at the centre of $P$, into $n_d = 2$ *dependent* angles $d$ and the remaining *free* angles $q$; any split is valid, and the checker takes the pair for which the $2 \times 2$ block of the differential of $C$ at $c_P$ has the largest determinant (a floating-point

choice that affects only efficiency). For a family without a constraint $n_d = 0$ and all parameters are free. The unknowns are $w = (\theta_2, \dots, \theta_6, d) \in W = X \times D$, $n = 5 + n_d$, and the equations are

$$G(w; q) = (F_2, \dots, F_6,\ (\operatorname{Re} C,\ \operatorname{Im} C)) \in \mathbb{R}^n$$

(the last two only if there is a constraint). Every solution $(\theta, p)$ with $p \in P \cap \Sigma$ is a zero $w^* \in W$ of $G(\cdot; q)$ for the corresponding $q \in Q$. Let $\tilde{w}$ be the centre of $W$, $c_Q$ that of $Q$, $\rho$ the (upward-rounded) half-widths of $W$, and $Y$ any real $n \times n$ matrix; the kernel takes a floating-point inverse of the Jacobian at $(\tilde{w}; c_Q)$ (and makes no exclusion if the inversion fails).

**Lemma 17.** Let $F_q(\tilde{w}; Q)$ be an interval matrix containing $\partial_q G(\tilde{w}; q)$ for all $q \in Q$, $J(W; c_Q)$ one containing $\partial_w G(w; c_Q)$ for all $w \in W$, and $J_q(W; Q)$ one containing $\partial_q \partial_w G(w; q)$ for all $(w, q) \in W \times Q$. Put

$$V = Y\, G(\tilde{w}; c_Q) + \sum_l \left(Y F_q(\tilde{w}; Q)\right)_{\cdot l}\, [-h_P, h_P],$$

$$\mathcal{M} = I - Y J(W; c_Q) - \sum_l \left(Y J_{q_l}(W; Q)\right)[-h_P, h_P],$$

$$\mathcal{K} = -V + \mathcal{M}\, [-\rho, \rho].$$

If for some $r$ the interval $\mathcal{K}_r$ is disjoint from $[-\rho_r, \rho_r]$, then $X \times P$ contains no solution.

*Proof.* Fix $q \in Q$ and suppose $G(w^*; q) = 0$ with $w^* \in W$. Applying the mean-value theorem to each component of $g(w) = w - YG(w; q)$ on the segment from $\tilde{w}$ to $w^*$ gives $w^* - \tilde{w} = g(w^*) - \tilde{w} = -YG(\tilde{w}; q) + (I - \Lambda)(w^* - \tilde{w})$, where the $r$-th row of $\Lambda$ is $\sum_s Y_{rs}\, \partial_w G_s(\xi_r; q)$ for some point $\xi_r$ of the segment, which lies in $W$. By the mean-value theorem in $q$, applied entrywise, $G(\tilde{w}; q) \in G(\tilde{w}; c_Q) + F_q(\tilde{w}; Q)(q - c_Q)$ and $\partial_w G(\xi; q) \in J(W; c_Q) + J_q(W; Q)(q - c_Q)$ for $\xi \in W$, with $|q_l - c_{Q,l}| \le h_P$. Because $Y$ is a point matrix, $Y(F_q(q - c_Q)) \subseteq \sum_l (YF_q)_{\cdot l}(q_l - c_{Q,l})$, and likewise $\Lambda \in YJ(W; c_Q) + \sum_l (YJ_{q_l})(q_l - c_{Q,l})$ entrywise; hence $w^* - \tilde{w} \in \mathcal{K}$. Also $w^* - \tilde{w} \in [-\rho, \rho]$. So $\mathcal{K}_r \cap [-\rho_r, \rho_r] \neq \emptyset$ for every $r$. □

The kernel evaluates all enclosures with Lemma 14 and all products and sums in outward-rounded interval arithmetic, applying $Y$ before multiplying by $[-h_P, h_P]$ so that the correlation between the equations captured by $Y$ is retained. The earlier version used a second-order Taylor form in $q$ for the value term; the first-order mean-value form above is simpler to verify and was sufficient.

### 6.4 Containment, coverage and the pair graphs

*Containment.* A box $[l, h]$ (per coordinate, grid units) lies in a hull $[\ell, \eta]$ if, for every coordinate with $\eta - \ell < 2^{16}$, the offset $o = (l - \ell) \bmod 2^{16} \in [0, 2^{16})$ satisfies $o + (h - l) \le \eta - \ell$. This integer test can only err by declaring a box not contained.

*Coverage verification.* For a certified parameter box $P$ with hull family $\mathcal{H}_P$ and root $P_7 \supseteq P$, the checker proceeds as follows.

1. Take the survivors of the root pass for $P_7$ at level 7 and apply the modulus test for $P$.
2. At the current level, discard every box contained in a hull of $\mathcal{H}_P$; apply the mean-value test and then the Krawczyk test to the others and discard the boxes they exclude.
3. If no box remains, report success. Otherwise, if the level is 12 or more than $3 \cdot 10^5$ boxes remain, report failure; else replace every remaining box by its 32 children, discard those excluded by the modulus test for $P$, and return to step 2.

**Proposition 18.** If the coverage verification reports success for $P$, then $\mathcal{U}(H(p)) \subseteq \bigcup \mathcal{H}_P$ for every $p \in P \cap \Sigma$ (for every $p \in P$ if there is no constraint).

*Proof.* Let $p \in P \cap \Sigma$ and $\theta \in \mathcal{U}(H(p))$. The root pass starts from the closed dyadic boxes of level 4, which cover the torus, and removes only boxes excluded for $P_7 \supseteq P$ (Lemma 15), so $\theta$ lies in a

surviving box, and at each subdivision in one of its children. A box is removed in step 2 only if it lies in a hull, or if it contains no solution for any parameter of $P \cap \Sigma$ (Lemmas 15–17). Hence, as long as $\theta$ is not in a hull, it lies in some box of the current list; when the list is empty, $\theta$ lies in a hull. □

*Pair intervals and graphs.* For hulls $\mathsf{h}, \mathsf{h}'$ with centres $c, c'$ and half-widths $n, n'$ (units), $\langle v(\theta), v(\phi)\rangle = 1 + \sum_{k=2}^{6} e^{i(\phi_k - \theta_k)}$ with $\phi_k - \theta_k \in [c'_k - c_k \mp (n_k + n'_k)]A$; the kernel encloses this sum in a rectangle $\mathcal{T}$ (Section 5.3) and $|\langle v, w\rangle|^2$ in the interval $\Pi = |\mathcal{T}|^2$. The edge $\mathsf{h} \sim_O \mathsf{h}'$ is set if $\inf \Pi \le 0$ and $\mathsf{h} \sim_B \mathsf{h}'$ if $6 \in \Pi$; the self-separation (Sep) is certified if, with half-widths $2n_k$ about the centre 0, $\inf \mathrm{Re}\, \mathcal{T} > 0$ and $\inf |\mathcal{T}|^2 > 6$.

**Lemma 19.** The graphs so constructed contain the graphs $O$, $B$ of Section 2.4 for the true pair intervals, and a hull certified by the self-separation test satisfies (Sep). □

Extra edges only make (Cl) harder to satisfy, so Lemma 19 is the direction needed. The checker enumerates all six-cliques of $O$ by a set-based recursive search and, for each, searches for a six-clique of $O$ inside the common $B$-neighbourhood of its vertices; (Cl) holds if none is found. Neither the enumeration nor the graphs share code with the generator, which uses bit-set enumeration on graphs built by different formulas (Section 7).

### 6.5 The cover tree

**Lemma 20.** Let the checker's selector (Sections 3.4 and 4.3) be applied from level 3: the kept boxes of level 3 form the first generation, and a kept box that is not certified is replaced by its kept children. If this process terminates, by level 12, with every remaining box certified and verified (Proposition 18 together with the checks of Lemma 19 and the clique search), then (O1)–(O4) of Theorem 7 hold.

*Proof.* Every $p \in \mathcal{F}$ has one half-open owner at each level, retained by Lemma 11 or Section 4.3. Its next owner is a child of its present owner, including grid boundaries and the torus seam. The path reaches a certified cell. Proposition 18 on its enclosure implies (O2) on the ownership cell; the remaining checks give (O3)–(O4). □

*Verification by parameter subdivision.* Obligation (O2) is needed only for $p \in \mathcal{F} \cap Q$, where $Q$ is the parent ownership cell and $P = \overline{Q}$. Each such point follows a retained descendant owner within $Q$. When coverage cannot be certified on all of $P$, the checker tests the same hull family on the closed enclosures of all retained descendants at levels $\ell + 1$, $\ell + 2$, or $\ell + 3$, up to level 12. Proposition 18 on these enclosures proves coverage on $\mathcal{F} \cap Q$. Hull-only (Sep) and (Cl) then establish (O2)–(O4). Boundary points owned by another parent are not claimed to follow this parent's descendants. Eight submitted certificates required $\ell + 3$.

## 7. The generator

The generator (`gen/`, about 1,000 lines of Python with NumPy) produces the certificates. None of its components is needed for the correctness of Theorems 1 and 3: a wrong hull family would be rejected by the checker, and a missing box would leave a hole that the checker's cover tree reports. The generator is nevertheless written to be sound in its own right, both because an unsound generator wastes time on certificates the checker rejects and because the referee's findings concerned it.

*Arithmetic.* The generator uses complex discs, NumPy products and a padding rule described in Appendix A. We withdraw the claimed general proof of that rule. The earlier counterexamples and 20,000 random disc products remain diagnostic regression tests, not universal assurance. The generator's Arb table check is likewise separate from the checker's fresh interval verification. The theorem does not depend on soundness of the generator arithmetic.

*Selector.* The generator's parameter selector implements the same three conditions as the checker's (Section 3.4) with its own formulas and disc arithmetic, and no library trigonometric function; it keeps a subset of the boxes kept by the earlier, non-rigorous selector (14,658 against 15,648 level-9 boxes), the difference being the additional half-plane $\operatorname{Re}\alpha \geq 0$ and the sharper remainder $6h^2$ in place of $12h^2 + 10^{-9}$.

*Per-box procedure.* For a root (a kept level-7 box) the generator runs its own modulus-only root pass (phase levels 4 to 7), locates the unbiased vectors at a parameter point of the root numerically (damped Newton from 20,000 random starts), and then treats each kept level-9 child $P$:

1. the root survivors are re-tested for $P$ with a modulus test, a mean-value test and a parametric Krawczyk test (second-order Taylor form in the free parameters);
2. the numerical solutions are re-converged at a point of $P \cap \Sigma$; each receives a *declared radius* derived from a Krawczyk interval at level $L_{\max}$, clamped to $[0.02, 0.15]$;
3. surviving boxes are subdivided and re-tested down to level $L_{\max}$ (9, and 10 in a second attempt); from level $L_d$ (7, respectively 8) on, boxes within the declared radius of a numerical solution are attached to it and not subdivided;
4. uncovered survivors are used as Newton starting points ("rescue", at most three rounds), and any still uncovered are grouped into connected clusters;
5. each solution with attached boxes, and each cluster, yields one hull: the smallest cyclic box containing all its boxes;
6. the self-separation and clique conditions are checked with the generator's own pair intervals and a bit-set clique enumeration.

Numerical solutions influence only the grouping of surviving boxes, never whether a box survives. Two defects found by the referee are repaired. The cyclic hull (step 5) now reduces every interval modulo $2^{16}$, splits wrapping intervals, merges the pieces and takes the complement of the largest gap including the wrap-around gap; the referee's example ($[1000, 1100]$ and $[65000, 68000]$) and 3,000 adversarial cases with wrapping and nested intervals are contained correctly (comment 4.4). Hulls are formed from the attached dyadic boxes only, not from declared balls, and the Krawczyk contraction of the earlier version was dropped: on 82 sample boxes it changed no hull size and no outcome. Records now carry an explicit state — *certified*, *refined* (the box was subdivided; its hulls are not stored) or *failed* — and on resumption only certified records are skipped (comment 4.5, with a regression test inserting the referee's synthetic record).

If a box fails at $L_{\max} = 10$ it is subdivided and its kept children are treated in the same way, down to parameter level 12. The certified records store the hulls on the grid $2\pi\mathbb{Z}/2^{16}$; these, and nothing else from the generator, are read by the checker.

# 8. Verification architecture

## 8.1 What the proof depends on

Figure 1 shows the flow of data and trust. Theorems 1 and 3 depend on the following, and on nothing else:

1. the mathematics of Sections 2–6 (Lemmas 4–20, Propositions 6, 10, 12, 18 and Theorem 7);
2. the exact computations that support Lemma 9, Proposition 10 and Proposition 12 (the SymPy script `exact/lemma5_exact.py` and the three scripts in `fourier/`, which use integer and rational arithmetic);
3. the checker: the kernel `chk/kernel.c` and Python modules `chk/ck.py`, `chk/run_check.py`, `chk/verify_split.py`, `chk/receipt_binding.py` and `chk/full_replay.py`, which build the family data from the defining formulas, implement the exact ordering test, drive branch and bound (including parameter subdivision), bind replay inputs, run the clique search and

walk the cover tree; current line counts and hashes are in Appendix C;

4. `mpmath` interval arithmetic, used for the table verification, the constants and the coefficient rectangles of the Fourier family;
5. the usual platform: IEEE-754 binary64 hardware arithmetic with correctly rounded $+, -, \times, \sqrt{}$; a C compiler that honours `-ffp-contract=off` and does not reorder floating-point operations at `-O2`; the Python interpreter and NumPy for integer array manipulation and control flow (no NumPy floating-point result enters an enclosure; the family data are exact integers and the kernel receives only integers and verified binary64 end-points).

The generator, its numerical solvers, its Arb check of the table, its selector, its pair intervals and its clique enumeration are *not* in this list. The checker reads the generator's results files as plain JSON data (box indices and hull end-points); it imports no generator module. The only artefact produced by the generator that the checker uses beyond the certificates themselves is the unit table, which the checker verifies independently with a different library (`mpmath` rather than Arb).

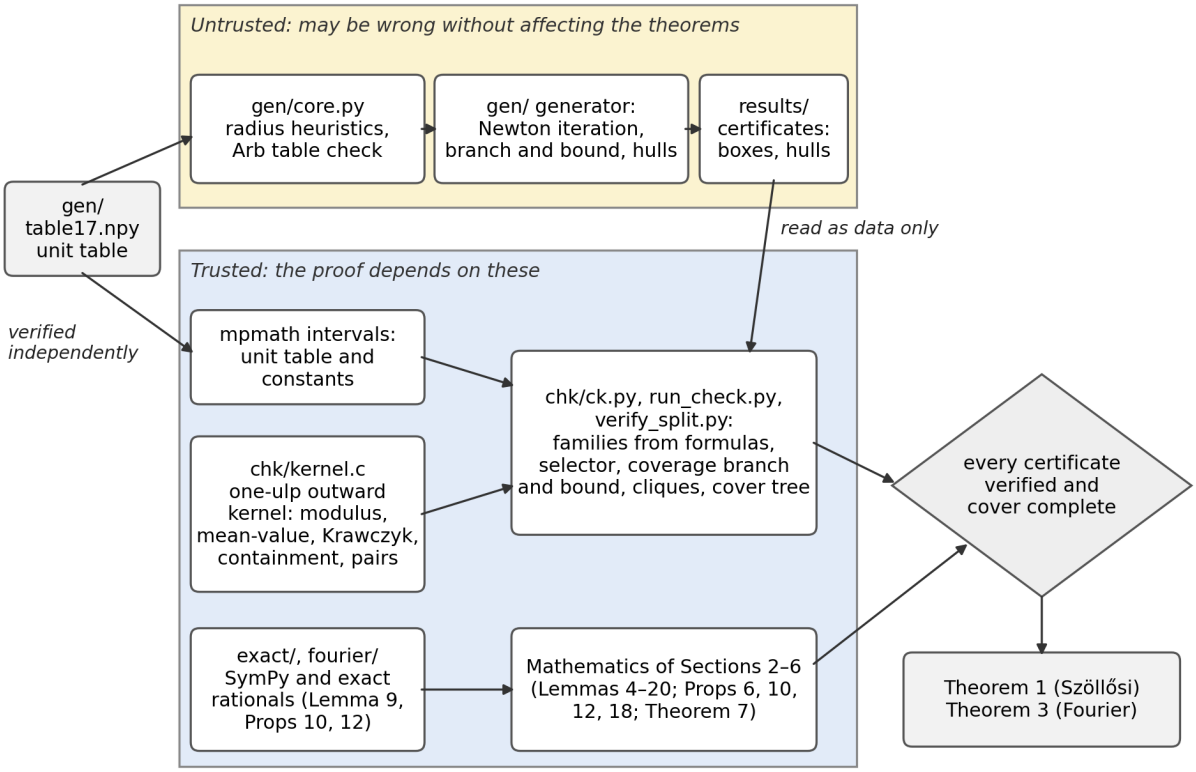



Figure 1: Data flow and trust. Theorems 1 and 3 depend only on the components in the lower (trusted) box; the generator may be wrong without affecting them, because every certificate it produces is re-verified by the checker.

## 8.2 Independence boundaries

The referee asked for a precise statement of what each verification step checks and what it shares with the main computation. Table 1 gives it.

| Question | Verified by | Shares with the generator |
|---|---|---|
| Is the unit table correct to $2^{-53}$ per component? | checker, `mpmath.iv`, 40 digits | the table file (the object being checked) |

| Question | Verified by | Shares with the generator |
|---|---|---|
| Does the selector keep the canonical owner of every point in $\mathcal{F}$? | checker's selector (Lemma 11, Section 4.3) | nothing (different formulas and arithmetic) |
| Is every kept box certified (cover completeness)? | checker's cover tree (Lemma 20) | the list of certified boxes (checked data) |
| Do the hulls contain all unbiased vectors? | checker's coverage B&B (Proposition 18) | the hull end-points (checked data) |
| Are the hulls self-separated, and is (Cl) true? | checker's pair intervals and set-based clique search | the hull end-points (checked data) |
| Are the symmetry identities and fundamental domains correct? | SymPy and exact rational scripts | nothing |

Table 1. What is verified independently of the generator.

Here "independent checker" means implementation separation from the generator. It does not mean independent human review or external reproduction of every certificate. The supplied 30 September review reports fresh graph checks for all certificates but fresh phase-space replay for seven selected parents only. Its remaining phase-space evidence was the submitted replay history.

**Replay binding (v3).** Each success now carries full SHA-256 binding to the family arrays, parameter indices and level, hull endpoints, unit table, checker and subdivision sources, build script, compiled kernel and runtime versions. Legacy or mismatched caches are rejected. `--clean` requires a new output and forbids partial-run options; proof drivers disallow `MUB6_KERNEL`. Content hashes bind inputs but do not authenticate a dishonest cache writer: the clean full replay imports no historical success records. Explicit runtime checks replace assertions on the proof path, including the family-installation call, so Python optimization cannot remove them.

The v3 regression suite retains the boundary-owner example, stale-cache driver attack, changed-input binding tests and an exact generic mixed-parameter Krawczyk fixture. The latter shows why dropping the mixed term can falsely exclude a genuine root; it is not claimed to kill the surviving mutation of the MUB kernel.

### 8.3 Tests

*Checker test-suite* (`chk/tests/test_kernel.py`, eleven groups, all passing on both machines): T1 compares interval addition, subtraction, multiplication, squaring and square roots with exact rational arithmetic (16,052 enclosures, including the referee's value $1 + 2^{-52}$, zeros, subnormals and the largest finite number); T2 checks 3,000 enclosures of $e^{i\varphi}$ over angle intervals up to three full turns against 40-digit values; T3 checks 10,800 rectangles for $S_j$ at random points of random phase and parameter boxes of both families; T4 submits 373,188 dyadic phase boxes containing numerically converged solutions (residual below $10^{-10}$) to the modulus, mean-value and Krawczyk tests, none of which may exclude them; T5 samples pair values and checks the self-separation bound and the orthogonality edges of the Fourier basis; T6 checks that the selector keeps the boxes of 5,465 (point, level) pairs of the fundamental domain, including $\alpha = 0$ and both edges of the sector; T7 compares the containment test with brute-force enumeration and with exact expectations for boxes whose end-points coincide with, or lie one grid unit beyond, those of a hull, including across the wrap-around (950 decisions); T8 compares the exact ordering test with brute-force enumeration over rational sub-grids; T9 compares the family data with the displayed formulas and checks the Hadamard property on the constraint surface; T10 stresses the three exclusion tests with coarse parameter boxes (levels 3–7) and non-dyadic phase boxes of widths 1

to 1024 grid units placed so that a solution lies just inside a face (1,385,847 boxes); T11 checks the Krawczyk assembly directly: for sampled vertices and interior points $(w, q)$, the exactly evaluated point $(w - \tilde{w}) - YG(w; q)$ must lie in the kernel's interval vector $\mathcal{K}$ (2,880 points).

*Mutation testing* (`chk/tests/mutate.py`; Jia and Harman 2011). The supplied record reports 16 of 17 kernel mutations detected by the test suite. The survivor removes the mixed-parameter Jacobian term in Lemma 17. This is incomplete test coverage, not evidence that the term is unnecessary or impossible to test. The exact generic fixture $G(w; q) = (1+q)w-5/2$, $W = [-1, 1]$, $Q = [0, 2]$, $\tilde{w} = 0$, $c_Q = 1$, $Y = 1/2$ has root $(w, q) = (5/6, 2)$. Omitting the mixed term gives $\mathcal{K} = \{5/4\}$, falsely disjoint from $W$; including it gives $[3/4, 7/4]$, containing the root. This permanent rational-arithmetic test is not a claim to have killed the specific MUB-kernel mutant.

*Generator regression tests* (`tests/test_gen.py`) reproduce each counterexample of the review and check the repair: the disc product and the point–interval matrix product for $r = 1+2^{-52}$ (also with zero left radius); 20,000 random disc products against exact rational arithmetic; the cyclic-hull example and 3,000 adversarial wrapping and nested cases; the referee's synthetic "refined" record, which is no longer counted as certified; and the selector on sampled points of $\mathcal{F}$ and on the special points $\alpha = 0$, $\alpha = 1$ and the ray.

*Historical platforms.* The submitted kernel test history records three platforms (Section 9.1): x86-64 Linux with GCC 13.3, ARM64 Linux with GCC 11.4, and ARM64 macOS with Apple clang 21. Those historical logs report all groups passing; on the two platforms where the suite was run in one piece (x86-64 Linux and macOS) every reported count is identical. They do not establish a three-platform replay of the revised Python drivers. The current local environment and fresh results are recorded separately in the release receipt. On ARM64, compilers fuse multiplications and additions by default; the explicit `-ffp-contract=off` prevents this.

# 9. The computations

This section preserves the submitted generation and replay history and its measured statistics. These historical machine runs are distinct from the revised clean replay described in the release receipt. Historical cache-based summaries are not by themselves verification of the revised driver; the new release requires fresh input-bound coverage checks.

## 9.1 Set-up

Three machines were used. *Machine A* is a cloud virtual machine (two Intel Xeon cores at 2.1 GHz, 8 GB, Ubuntu 24.04, Python 3.11, NumPy 2.4.4 with OpenBLAS, `mpmath` 1.3.0, `python-flint` 0.9.0, GCC 13.3). *Machine B* is an Apple M2 Mac mini (eight cores, 16 GB, macOS, Python 3.14, NumPy 2.4.2, `mpmath` 1.3.0, `python-flint` 0.9.0, Apple clang 21), shared with other work during the run. *Machine C* is an ARM64 Linux virtual machine on the same Mac (Ubuntu 22.04, Python 3.10, NumPy 2.2.6, GCC 11.4), used for the test-suite. BLAS threading was disabled in every process. The unit table used by all runs has SHA-256 `075160008692331ecec57ebd851514642c5800b39d9bcad11c668c01074c5969`; it was verified by the generator with Arb and by the checker with `mpmath`, independently, on each machine.

The Szöllősi roots (kept level-7 boxes) were ordered by decreasing $|\alpha|$ at their centres. Machine A generated the certificates for the roots at positions $\equiv 6 \pmod 7$ (195 roots, two processes); machine B generated those at the other positions (six processes), except the last 250 of its list, which machine A generated after finishing its share. The Fourier certificates were generated on machine A. The checker was run on machines A and B, each replaying part of the certificates, and the final cover evaluation was run on the union of all checker outputs.

### 9.2 The Szöllősi family

The checker's selector keeps 160, 183, 232, 532, 1,367, 4,294 and 14,658 parameter boxes at levels $3, \dots, 9$; the 14,658 kept level-9 boxes lie below 1,088 of the 1,367 roots (the other roots have no kept child). The generator certified every one of the 14,658 level-9 boxes at level 9 (12 of them in the second attempt with $L_{\max} = 10$); the only failed records are those of the incident described in Section 9.4.

*Historical checker replay.* The submitted checker history reports all 14,658 certificates, 75 of them by parameter subdivision (Section 6.5) at parameter levels 10 to 12. Its cover evaluation, run on the union of its outputs, reports every kept box of level 3 covered (`chk_results/chk_szollosi_summary.json`). For the revised release, this historical summary is not the acceptance gate: Theorem 1 additionally requires the fresh input-bound full replay recorded in `REPLAY_RECEIPT.md`, together with the semantic arguments of Theorem 7, Lemma 20 and Proposition 10.

The 75 boxes that needed parameter subdivision all contain parameters with $z = w = 1$ ($\theta_x + \theta_y \equiv \theta_u + \theta_v \equiv 0$), where $\alpha = 1 + 2\cos\theta_x$ is real: they lie on the edge $\arg\alpha = 0$ of the sector, with $\alpha$ between about 0.16 and the Diţă point $\alpha = 1$, and $\theta_z, \theta_w$ sit at the wrap-around point $0 \equiv 2\pi$ of the ordering conditions. For 2 of them no kept sub-box exists at the next level (the box does not meet $\mathcal{F}$), so the verification is vacuous. For the others the difficulty lies with the checker's first-order parametric Krawczyk test on the full parameter box, not with the hulls: a numerical probe, which is not part of the proof, sampled parameters on $\Sigma$ in 4 of these boxes and found all 60,864 numerically computed unbiased vectors inside the stored hulls (`analysis/probe_split_boxes.log`).

Table 2 summarises the certificates. The number of numerically located solutions, which only steers the grouping of boxes into hulls and is not a certified count, is given as a complete histogram.

| quantity | value |
|---|---|
| certified level-9 boxes (generator), replayed and verified (checker) | 14,658 / 14,658 |
| numerically located solutions per box (complete histogram, value: boxes) | 42: 3; 48: 17; 49: 1; 54: 567; 55: 10; 56: 7; 57: 6; 58: 1; 59: 1; 60: 12,817; 63: 4; 64: 4; 72: 768; 84: 125; 96: 62; 108: 48; 120: 217 |
| boxes with rescued solutions; solutions rescued | 448; 5,229 |
| boxes with extra hulls from left-over clusters; extra hulls | 7,310; 40,247 |
| hulls per box: minimum, median, maximum | 0, 61, 141 |
| boxes without hulls | 2 |
| largest hull half-width per box: median, 99th percentile, maximum | 0.147, 0.233, 0.430 |
| six-cliques of $O$ (checker): median and maximum per box; total | 9, 24303; 2,765,699 |
| edges of $O$ and $B$ (checker), maxima | 1,391, 4,493 |
| root-pass survivors at level 7 (checker): minimum, median, maximum | 1,778,718, 2,303,380, 7,177,859 |
| CPU time: generator, checker (hours) | 41.5, 15.4 |

Table 2. Statistics of the Szöllősi certificates (generator) and of their replay (checker).

The checker's coverage branch and bound needed phase levels up to 12 (the deepest level needed was 7 for 8 boxes, 8 for 2,407 boxes, 9 for 11,188 boxes, 10 for 1,040 boxes, 11 for 12 boxes, 12 for 1 box). The total generator time was 42 CPU-hours and the checker time 15 CPU-hours.

*Special parameters.* At the special parameters: the box (0, 170, 0, 170) containing $\alpha = 0$ was verified directly, with 49 hulls (60 numerical solutions, largest hull half-width 0.172); its graph $O$ has 212 edges and 60 six-cliques and $B$ has 451 edges; the box (19, 137, 203, 322) containing the ray point $\alpha = \frac{1}{2} + \frac{i}{2\sqrt{3}}$ was verified directly, with 54 hulls (54 numerical solutions, largest hull half-width 0.160); its graph $O$ has 112 edges and 1 six-clique and $B$ has 336 edges; the box (0, 128, 0, 256) containing the Diţă point $\alpha = 1$ was verified directly, with 120 hulls (120 numerical solutions, largest hull half-width 0.098); its graph $O$ has 826 edges and 10 six-cliques and $B$ has 2,710 edges. In each case no six-clique of $O$ has a partner clique across $B$.

### 9.3 The Fourier family

The selector of Section 4.3 keeps 3, 5, 15, 41, 143, 505 and 1,935 boxes at levels $3, \dots, 9$. The generator certified all 1,935 level-9 boxes at level 9, and the checker verified all of them; its cover evaluation is complete (`chk_results/chk_fourier_summary.json`). By Theorem 7 and Proposition 12 this proves Theorem 3.

The certificates agree with the numerical observations of Jaming et al., which they do not depend on. In every box the generator located 48 unbiased vectors numerically (their Numerical Evidence 1.3: 48 for all $(a, b)$). The six-cliques of the graph $O$ are the candidate orthonormal bases: in 1,876 of the 1,935 boxes there are exactly 8 of them (their observation for generic parameters), in the box at $(a, b) = (0, 0)$ there are 16 (they report 16 bases for $F_6$), and in the box containing $(a, b) = (1/6, 0)$ there are exactly 70 (they report 70). The largest number, 897, occurs in the neighbouring box (84, 0), where hulls of nearby vectors overlap.

| quantity | value |
|---|---|
| certified level-9 boxes (generator), replayed and verified (checker) | 1,935 / 1,935 |
| numerically located solutions per box | 48: 1,935 |
| hulls per box: minimum, median, maximum | 48, 48, 55 |
| largest hull half-width per box: median, maximum | 0.049, 0.172 |
| six-cliques of $O$ (checker): median and maximum per box | 8, 897 |
| deepest phase level in the checker (level: boxes) | 8: 74; 9: 1,858; 10: 3 |
| CPU time: generator, checker (hours) | 0.48, 0.38 |

Table 3. Statistics of the Fourier certificates and their historical replay.

### 9.4 History of the run

Three incidents are recorded because the raw results files contain their traces; none of them touches the checker, and every record written in these circumstances is either a certificate, which the checker verifies like any other, or a failed or refined record, which is not used.

First, machine B's Python installation lacked SciPy, which the generator used for the connected components that group left-over boxes into extra hulls. The resulting exceptions were treated, correctly, as failed certificates, and the affected level-9 boxes were subdivided down to level 12, where 860 records ended in the state *failed*. The component routine was replaced by a SciPy-free union–find in NumPy (tested against SciPy on 300 random graphs), the run was resumed, and every affected level-9 box was then certified at level 9; the checker's cover tree stops at those certificates.

Second, machine B's generator was restarted twice: once with a limit on its list of roots, to hand the last 250 roots to machine A, and once for the repair just described. The resumption logic skipped every box already certified. On macOS the worker processes of an interrupted run are not terminated with their parent, and those of one interrupted run continued for about half an hour, appending records, before they were found and stopped.

Third, machine A's container was restarted by its host during the tail run, which then resumed from its results file.

### 9.5 Relation to the earlier computation

The earlier computation certified 15,648 level-9 boxes selected by the non-rigorous selector; the rigorous selector keeps 14,658, all of them among the earlier ones. None of the earlier certificates is used. As a test of the checker's strength on the hardest parameters, it was also applied to six earlier certificates near $\alpha = 0$, on the ray and near $\alpha = 1$, and verified all of them.

## 10. Numerical diagnostics of degenerate parameters

This section reports numerical observations that are **not part of the proof**; the certificate needs neither their correctness nor any statement about the structure of $\mathcal{U}(H)$. The earlier version described coalescing unbiased vectors on the ray $\arg\alpha = \pi/6$ and at $\alpha = 0$ as coinciding "exactly" and as not isolated; the referee pointed out that nearby clustering and small singular values do not distinguish an isolated multiple root from a positive-dimensional component. The observations below are therefore stated as diagnostics, computed in 60-digit arithmetic (`exact/degeneracy_diagnostics.py`): solutions were located by multistart Newton iteration in binary64 (40,000 starts), refined by damped Gauss–Newton iteration in `mpmath` until the residual of all six equations was below $10^{-40}$, and de-duplicated at distance $10^{-18}$; for each refined solution the singular values of the $6 \times 5$ real Jacobian were computed. A singular value below $10^{-20}$ is reported as "numerically singular".

| parameter $\alpha$ | distinct solutions | numerically singular | smallest singular value (others) | minimal distance | clusters (linkage $\le 10\delta$) |
|---|---|---|---|---|---|
| $0.7 + 0.1i$ (generic) | 60 | 0 | 0.84 | 0.33 | — |
| 1 (boundary, Diţă) | 120 | 0 | 1.11 | 0.71 | — |
| ray, $\delta = 10^{-2}$ | 60 | 0 | $1.9 \cdot 10^{-2}$ | $7.5 \cdot 10^{-3}$ | two of size 4 |
| ray, $\delta = 10^{-4}$ | 60 | 0 | $1.9 \cdot 10^{-4}$ | $7.5 \cdot 10^{-5}$ | two of size 4 |
| $\frac{1}{2} + \frac{i}{2\sqrt{3}}$ (on the ray) | 54 | 2 | 1.25 | 0.67 | — |
| near 0, $\delta = 10^{-2}$ | 60 | 0 | $9.0 \cdot 10^{-3}$ | $3.45 \cdot 10^{-3}$ | six of size 4 |
| near 0, $\delta = 10^{-4}$ | 60 | 0 | $9.0 \cdot 10^{-5}$ | $3.45 \cdot 10^{-5}$ | six of size 4 |
| 0 | 42 | 6 | 3.0 | 1.32 | — |

Table 4. Unbiased vectors at special parameters. Distances are $\max_k |e^{i\theta_k} - e^{i\phi_k}|$; clusters are

single-linkage clusters at that distance. "Ray, $\delta$" is $\alpha = 3^{-1/2}e^{i(\pi/6-\delta)}$ and "near 0, $\delta$" is $\alpha = \delta e^{i\pi/12}$. The exact point of the ray is $\alpha = \frac{1}{2} + \frac{i}{2\sqrt{3}} = 3^{-1/2}e^{i\pi/6}$ (the earlier version gave the decimal approximation $0.5 + 0.2887i$).

The pattern is consistent across the table. Approaching the ray, two groups of four solutions contract linearly in $\delta$, and the smallest singular value of the Jacobian at their members decreases linearly as well; at the exact ray point 54 distinct solutions are found, 52 with well-conditioned Jacobians and 2 at which the Jacobian is numerically singular, which is what one expects if each group of four converges to a single singular solution ($52 + 2 \cdot 4 = 60$). Approaching $\alpha = 0$, six groups of four contract in the same way, and at $\alpha = 0$ there are $42 = 36+6$ distinct solutions, 6 of them numerically singular. At the smallest separations tried ($\delta = 10^{-6}$) the binary64 multistart no longer resolves all members of the groups (it finds 56 and 50 solutions respectively), which illustrates the difficulty faced by a certificate that isolates simple roots: at any fixed resolution there are parameters near these loci at which members of a group are closer together than the resolution, and a simple-root isolation test (such as the Krawczyk test used here, applied to single boxes) cannot succeed on a box containing two of them. The present certificate avoids the question by enclosing whole groups in single hulls. Whether the singular solutions are isolated multiple roots or points of positive-dimensional components is not decided by these computations and is not needed.

# 11. Discussion

## 11.1 What has been proved

Theorem 1 settles Conjecture 3 of MMVW for every complex Hadamard matrix of the two-circulant form (4.2), that is, for every $\alpha$ in the closed region $\mathbb{D}$, boundary and degenerate loci included. With Theorem 3 (Jaming et al.'s result, re-proved here), both families that appear in MMVW's Proposition 4.2 are now excluded by certificates verified by the same checker, so that MMVW's route to Zauner's conjecture in dimension six depends only on their Conjecture 2. Theorem 1 says nothing about Conjecture 2, nor about complex Hadamard matrices outside the two families. The classification of complex Hadamard matrices of order six announced by Cárdenes Wuttig and Tindall (2026) is a statement about Hadamard matrices, not about mutually unbiased bases, and is not used.

## 11.2 Comparison with Jaming et al. (2009)

Jaming et al. replaced the parameters by the centres of $N = 180$ cells per unit of $a$ and $b$ (270 cells for the fundamental triangle) and the phases of candidate vectors by the centres of $N' = 19$ cells per turn, refined dyadically for seven generations. Their Lemma 3.1 (the "trivial error bound") and Lemma 3.2 (the maximal deviation of an inner-product sum over a product of small intervals is attained at end-points) give explicit conditions that a cell must satisfy if it contains a vector unbiased to $F(a, b)$ or orthogonal to another; an exhaustive search over the surviving "quasi-unbiased" vectors then showed that no two "$N'$-orthonormal" bases that are "$N'$-unbiased" to each other exist. The program ran for about six hours. The argument has the same three components as ours — a parameter cover, an enclosure of the unbiased vectors per cell, and a combinatorial exclusion — and MMVW rightly regarded it as the template for Conjecture 3. The present work differs in the following respects, which are the improvements it claims.

- *Rounding.* This package supplies a separate outward-rounded checker, a verified unit table and explicitly checked constants. We do not make an implementation-specific criticism of the historical program without an archived source snapshot. The published 2009 mathematical theorem remains the predecessor being re-proved, not an unproved claim.
- *Exclusion strength.* Their cell tests are of order zero: they bound how far the inner-product

sums can move within a cell, without derivative information, so that a cell can be discarded only once it is small compared with its distance from the solution set. Ours add a mean-value test and a parametric Krawczyk test, both using derivatives with respect to the phases and the parameters, so that for parameter cells of width $2\pi/2^9$ the phase-space subdivision stops at level 9 or 10 even close to solutions.
- *Clusters.* Near degenerate parameters the unbiased vectors of the Szöllősi family coalesce (Section 10). The certificate does not need to separate them: a hull may contain a cluster, and the self-separation condition is all that the combinatorial step requires. The Fourier family has no such loci in its triangle, which is consistent with the absence of any special treatment in Jaming et al.
- *Verification design.* The certificates are produced by one program and verified by another, with a small trusted kernel and an explicit list of what the proof depends on (Section 8).
- *Transferability.* The Fourier family was handled by the same generator and checker as the Szöllősi family; the family-specific inputs were the exponent data, the coefficient rectangles and the selector.

### 11.3 Assumptions and limits

The trusted components are listed in Section 8.1. The mathematics is not formalised in a proof assistant. Mutation coverage is incomplete: one supplied kernel mutation survived. Lemma 17 and the exact generic assembly fixture demonstrate the mixed term's necessity, but the fixture is not a kernel-specific mutation kill. The certificate does not assume that numerical root discoveries are complete or that $\mathcal{U}(H)$ is finite. Exact identities and finite checks retain their stated scope; neither producer-side replay nor model review establishes external peer review.

### 11.4 Transfer to other families

The implemented interface accepts entries that are unimodular constants times torus characters, with the specified trigonometric constraint structure. Transfer has been demonstrated only for the two families in this paper and their equivalences. Karlsson's three-parameter family and Szöllősi's four-parameter family are research directions, not established inputs to this checker. Their algebraic reconstruction, denominators, branch choices and exceptional cases require complete encodings and coverage proofs. The present kernel has capacities `MAXP=4`, `MAXT=8` and `MAXN=7`; an enlarged encoding could require kernel changes and renewed verification. Parameter counts alone do not prove compatibility, and no exclusion for either larger family is claimed.

### 11.5 Availability and disclosure

The release package provides the source, certificates, checker, replay records and a machine-oriented index. The scholarly attribution is Anonymous. The supplied manuscript credited Claude; Codex performed the present revision and publication checks. These are reported system roles, not externally audited authorship or human peer review. No human mathematical contribution is inferred from account ownership. Funding and conflicts were not recorded in the supplied research provenance; none are inferred. This is mathematical research without human-subject data. Original research contents are CC0-1.0 and original code is MIT, with third-party exceptions listed in `LICENSES.md`.

## Appendix A. Withdrawn generator rounding claim

The generator is outside the trusted computing base. The former Lemma R and its proof are withdrawn: multiplication combines uncertainty from both operands, invalidating the claimed intermediate induction. With $u = 2^{-53}$, two exact inputs $(1-u)^{-8}$ represented by computed values 1 have product $(1-u)^{-16}$, exceeding the asserted exponent 9 even with its small additive

allowance. This does not refute the final padded bound, but that unused bound is no longer claimed. The padding rule below remains a generator heuristic; every proposal must be checked independently.

The implemented rule is $\mathrm{up}(x) = \mathrm{fl}(\mathrm{fl}(x(1 + 2^{-40})) + 2^{-800})$.

Generator regression tests are diagnostics, not proof of a general rounding lemma. The theorem relies on the separate outward-rounded checker, not these heuristics.

Every radius returned by a generator primitive is passed through up (and replaced by $+\infty$ if it exceeds $2^{64}$ or is not a number); differences and lower bounds are formed by a single correctly rounded operation on exact operands followed by a one-ulp step, as in Lemma 13; and all exclusion decisions are strict comparisons of a lower bound with an upper bound, which are false whenever a NaN or infinity occurs.

## Appendix B. Response to the second review

This revision responds to the 30 September 2026 review of version 2. Historical logs remain evidence of the submitted version, not fresh verification of the revised driver.

| Obligation | Disposition |
|---|---|
| Closed enclosures versus half-open ownership | Theorem 7, Lemma 11 and Lemma 20 now use canonical half-open owners. Direct interval tests use closed enclosures; subdivision covers the retained fundamental-domain portion. The explicit boundary counterexample is retained as a regression. |
| Cache accepts changed hulls | Receipts bind the actual hulls, family, table, checker sources, compiled kernel and runtime. Legacy and mismatched receipts are rejected. Clean replay starts with no receipts; proof drivers reject kernel overrides. The stale-cache attack is an executable negative control. |
| Python optimization | A further defect found during revision placed the family-installation call inside an assertion. This is now an explicit runtime call; optimized-Python regression tests exercise it. |
| Unsupported larger-family transfer | Section 11.4 treats larger Karlsson and Szöllősi families as future work, with finite torus-character capacities and additional algebraic-branch obligations. |
| Invalid generator-rounding induction | Appendix A withdraws the unused lemma and explains the exponent error. Generator padding is not part of the proof; checker containment remains necessary. |
| Mixed-parameter Krawczyk term | An exact generic example demonstrates false exclusion if the term is omitted. This is not a claim to kill the surviving MUB-kernel mutation. |
| Subdivision depth | The account includes all eight historical depth-three parents and the maximum permitted level 12. |

| Obligation | Disposition |
|---|---|
| Historical numerical-code comparison | Unsupported implementation criticism has been removed. The Fourier precedent is credited as an existing theorem, not a new result. |
| Assurance and counts | Numerical solution counts, hull counts, graph checks and phase-space replay are distinguished. The supplied reviewer checked all graphs but only selected phase-space parents afresh. Implementation separation is not external peer review. |
| Reusable interface | A standalone certificate-interface description states ownership, arithmetic, coverage and graph obligations. Larger-family encoding and formalization remain optional future work, not achievements of this package. |

The release replay receipt, rather than historical per-machine summaries, determines the verification status of this revision.

## Appendix C. Inventory of code and data

Line counts and SHA-256 digests (first 16 hexadecimal digits) of the files on which the results depend; the bundle file `SHA256SUMS` lists the full digests of every file, including all results, checker outputs and logs.

**Checker (trusted)**

| file | lines | SHA-256 (prefix) |
|---|---|---|
| `chk/kernel.c` | 638 | `9214d0d935ad21a4` |
| `chk/ck.py` | 385 | `b134fbef3bae0b90` |
| `chk/run_check.py` | 162 | `382b7910b5195adc` |
| `chk/verify_split.py` | 77 | `8ba748ea3e78101e` |
| `chk/receipt_binding.py` | 58 | `45ade521a0d9173d` |
| `chk/full_replay.py` | 55 | `556b31cae41a8769` |
| `chk/build.sh` | 11 | `fcdb7df0fe374ff1` |

**Exact verification (trusted)**

| file | lines | SHA-256 (prefix) |
|---|---|---|
| `exact/lemma5_exact.py` | 111 | `e8142d5eb38c242f` |
| `fourier/symmetries.py` | 129 | `837092d6674b4a0a` |
| `fourier/fundamental_domain.py` | 111 | `1913323ee3ebb240` |
| `fourier/fourier_exact.py` | 17 | `6c64edd26c29f8c2` |

**Tests**

| file | lines | SHA-256 (prefix) |
|---|---|---|
| chk/tests/test_kernel.py | 408 | 1d55f31c282ccb07 |
| chk/tests/test_revision.py | 63 | 83dfeefc5abcdd6b |
| chk/tests/test_boundary_interval.py | 40 | 67470e36835c335d |
| chk/tests/mutate.py | 57 | 6f504b9dfbb59629 |
| tests/test_gen.py | 91 | 5806d6e653e7ffec |

**Generator (untrusted)**

| file | lines | SHA-256 (prefix) |
|---|---|---|
| gen/rnd.py | 63 | b5b23bec4e1cecff |
| gen/family.py | 87 | ceca8f8526730917 |
| gen/core.py | 320 | 45e90739fa9de956 |
| gen/hulls.py | 116 | 788317275e06297f |
| gen/comb.py | 73 | bff994877171e98f |
| gen/selector.py | 104 | eed4a8ade8ea638b |
| gen/solve.py | 197 | c692cdcf00c7fad7 |
| gen/driver.py | 131 | cd4fd20bdb3215b3 |
| gen/helpers_v2.py | 12 | ca1ba100a1baec90 |

**Diagnostics and analysis (not part of the proof)**

| file | lines | SHA-256 (prefix) |
|---|---|---|
| exact/degeneracy_diagnostics.py | 145 | 0169911f82c81913 |
| analysis/stats.py | 102 | c64b1811f6ac85d9 |
| analysis/empty_boxes.py | 31 | 6663ceeddaa6f7bf |
| analysis/ready_roots.py | 23 | c454da242201d947 |
| analysis/probe_split_boxes.py | 44 | 0f4c9d23bf6ea7cc |
| analysis/special_boxes.py | 23 | 082156ef59bb789f |

**Data**

| file | lines | SHA-256 (prefix) |
|---|---|---|
| gen/table17.npy | — | 075160008692331e |

Totals: checker (trusted), 1386 lines; exact verification (trusted), 368 lines; tests, 659 lines; generator (untrusted), 1103 lines; diagnostics and analysis (not part of the proof), 368 lines.

*Version boundary.* These hashes describe revised source. Historical logs refer to earlier versions and are not fresh receipts for these bytes. Revised clean-replay summaries bind their exact sources, compiled kernel, certificate inputs and environment. Generator variations cannot replace checker verification.

## Sources

Sources are grouped by theme; URLs are given for every item.

**The conjecture and its context**

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**Review of the earlier version**

Anonymous referee. (2026, September 29). *Review: Szöllősi-family MUB exclusion* [Unpublished referee report of the earlier bundle `mub6_szollosi_conjecture3.zip` (SHA-256 prefix 944044376cda648e), with the reproducible audit bundle `mub6_referee_audit_2026-09-29.zip` (SHA-256 prefix a4d08188e57262eb)]. Both are included in the bundle accompanying this paper (directory `review/`).